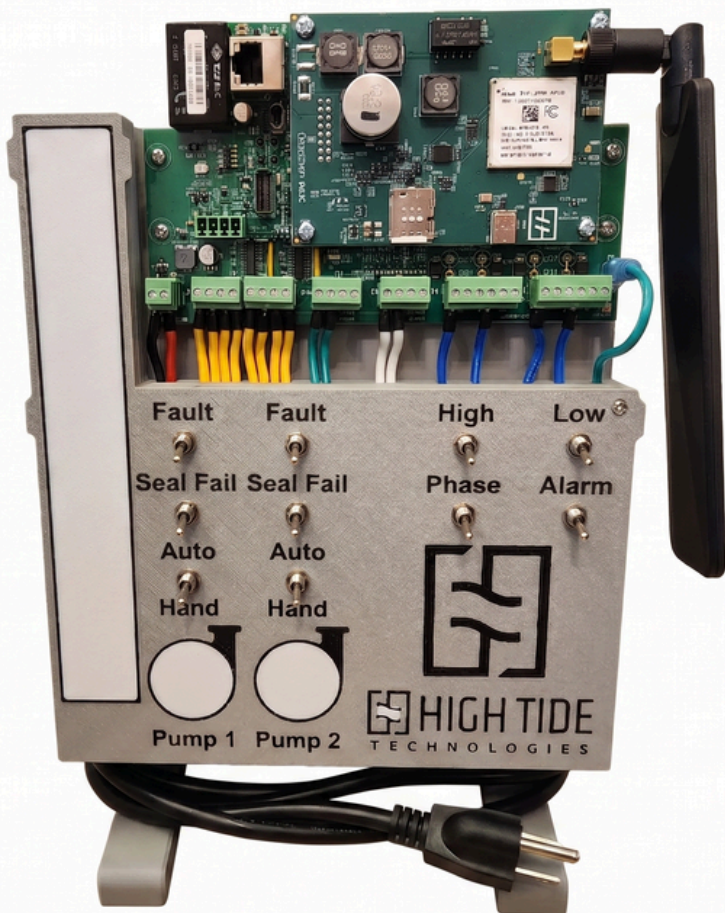




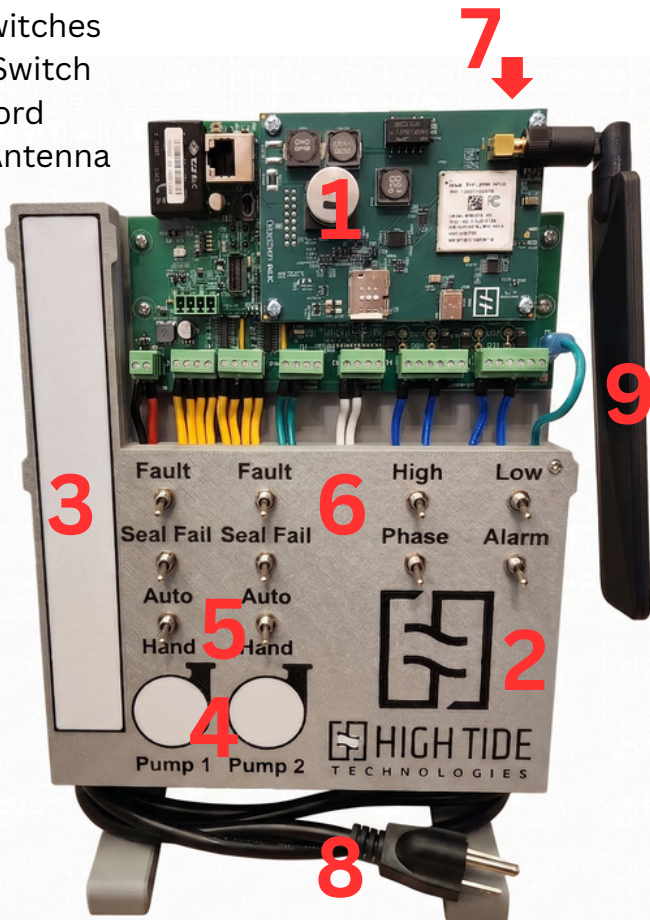
HIGH TIDE DEMO CASE

Operation Manual

DEMO CASE

Components Outside

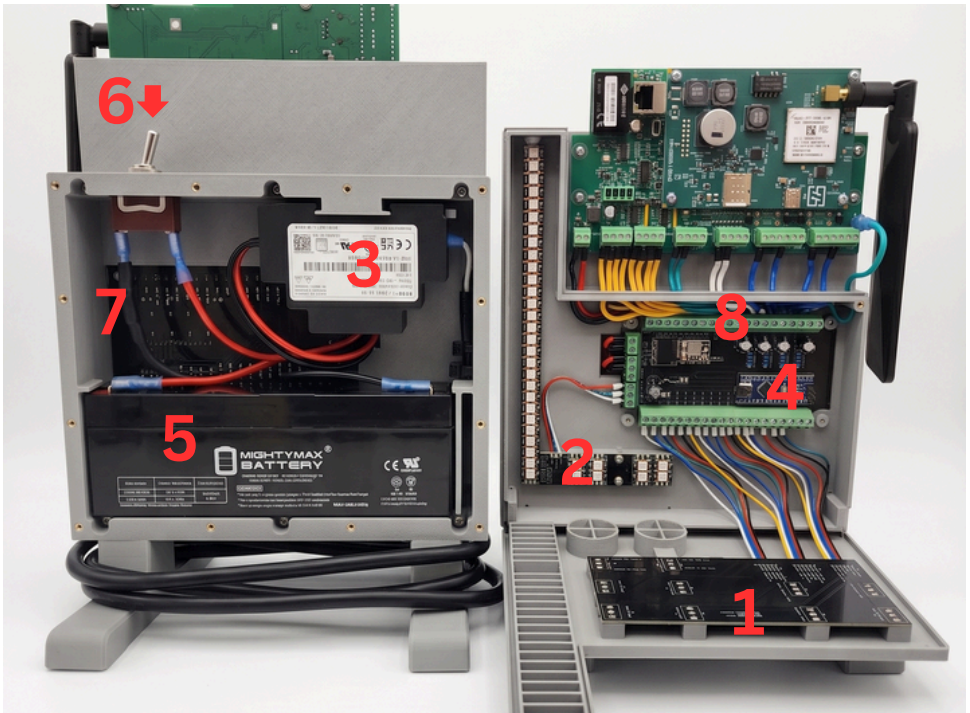
1. High Tide Gen-2 Board
2. Demo Case Assembly
3. Tank level indicator
4. Pump indicators
5. Pump HOA switch
6. Alarm Switches
7. Battery Switch
8. Power Cord
9. Flip Up Antenna



DEMO CASE

Components Inside

1. Switch Board
2. LED Board / Tank Level and Pumps
3. Power Supply
4. Artificial System Board
5. Battery
6. Battery Switch
7. Slow-Charge (resistor diode)



How To Use Setup

Your demo case is here to show off the capability of the High Tide Gen 2 board and how no one on the market can beat it's capabilities with a standard board.

Site should be setup like "Whitney Demo" site.

Inputs/Outputs

Digital Inputs

1. Pump 1 Seal Fail (NC)
2. Pump 2 Seal Fail (NC)
3. Pump 1 Fault (NC)
4. Pump 1 Fault (NC)
5. High Float Alarm (NO)
6. Low Float Alarm (NO)
7. Phase Fail (NO)
8. System Alarm (NO)

Digital Outputs

1. Pump 1 Call
2. Pump 2 Call
3. Unused
4. Unused

Digital Over Analog

3. Pump 1 Running
4. Pump 2 Running

Analog Inputs 0-4.2vdc (no resistor)

1. Tank Level ----- 0-100ft
2. System Flow ----- 0-1000gpm
3. Pump 1 Amps ----- 0-100amps
4. Pump 2 Amps ----- 0-100amps

Totals

10 Digital Inputs, 4 Analogs,
4 Outputs (only 2 used)

No one else can do this with their system.

Running a Demo or a Trade-Show

1. Powering On the Unit
 - a. Turn Battery Switch to ON.
 - b. Plug in the 120v cord.
2. Flip antenna UP.
3. Turn all alarms OFF (switch down).
4. Put pumps in AUTO (switch up).
5. Wait for the unit to connect.
 - a. If no connection, make sure the antenna is screwed in all the way.
6. Set up your demo site with your phone number or text so you can show how quickly alarms come in.
7. Leave the unit plugged in, and the battery switch on at all times to keep the battery charged and the system running in AUTO.

What the System is Doing

The Onboard Artificial System is acting like a water system. There is a tank, 2 pumps and flow meter. If the board is powered, it will start with the tank at 50% and you will see the system draining. The drain rate is random and pulls water like a regular drinking water system. If left alone, it will drain until there is nothing left in the tank. To fill the system, turn on one or both of the pumps. When they are running you will see a green spinning indicator. If the flow is more than the drain rate, the tank will fill. The pumps also output Amps to the board and the flow rate of the pumps. You can tell if the tank is losing or gaining water by the animation of the tank. The Pump indicators will change color depending on if they are running (Green), have a seal-fail (Purple), or are in fault (Flashing Red).

All these settings are made to simulate real data. For the best experience, keep the unit powered on all the time so you have data in the system to show off to the customer. This acts like a real system so when the pumps are in (AUTO), High Tide servers are calling the pumps to run. You can set any START/STOP points to your preferences.

Power and Trouble Shooting

1. The battery switch fully isolates the battery. You need this switch in (ON) for the battery to charge.
2. The Power Supply is set to 13.8v dc to float charge the battery.
3. The incoming power should be 120vac.
4. If the lights don't come on after you turn on the system:
 - a. Check that the 120v supply is good.
 - b. Battery could be dead; charge for 12 hrs if fully dead.
 - c. Tank is empty and pumps are off, switch a pump to (HAND) and see if the pump turns on.
 - d. If none of these work you may have a loose wire or a problem. Call us for trouble shooting.
5. If the unit is not connecting or turning pumps on automatically:
 - a. Check the pumps are in AUTO.
 - b. Make sure the Antenna is screwed in.
 - c. Check that you have service.

Programing

1. The Basic program can be found here on git hub:
 - a. <https://github.com/amku37/HighTide-Demo-Case>
 - b. Upload new software with your preferred settings by using Arduino IDE.
 - c. The notes will walk you through what each setting does. A USB-C to USB-A data cable is needed to upload software.

Demo Tips

Turn the unit ON and keep the demo unit charged so it's ready to go. When on battery mode, it has hours of run time. For short displays you don't need to plug it in. Make sure to turn OFF the battery switch when not in use, otherwise it will fully drain the battery. Lean into the HighTide Web interface to show how quickly we get alarms and what they look like for the customer. Also, show customers how the the graphs display. This works best if the demo unit is kept powered, so you can have data to look at and review in real time.